

XIANZHEN LUO

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EDUCATION

Harbin Institute of Technology

Ph.D. student in Code Intelligence

Supervisor: Prof. Wanxiang Che and Assoc. Prof. Qingfu Zhu

• **Honors:** National Scholarship

Harbin, China

Sep 2022 – Present

Harbin Engineering University

Bachelor in Computer Science

• **Honors:** National Scholarship, 2 x ICPC Regional Silver Medals, 2 x ICPC EC-Final Bronze Medals.

Harbin, China

Sep 2018 – Jun 2022

PUBLICATIONS

ICLR 2026 How Many Code and Test Cases Are Enough? Evaluating Test Cases Generation from a Binary-Matrix Perspective

Xianzhen Luo*, Jinyang Huang*, Wenzhen Zheng, Qingfu Zhu, Mingzheng Xu, Yiheng Xu, Yuantao Fan, Libo Qin, Wanxiang Che

ACL2025 **Outstanding Paper** Turning Trash into Treasure: Accelerating Inference of Large Language Models with Token Recycling

Xianzhen Luo, Yixuan Wang, Qingfu Zhu, Zhiming Zhang, Xuanyu Zhang, Qing Yang, Dongliang Xu

ACL2025 ChartCoder: Advancing Multimodal Large Language Model for Chart-to-Code Generation

Xuanle Zhao*, Xianzhen Luo*, Qi Shi, Chi Chen, Shuo Wang, Wanxiang Che, Zhiyuan Liu, Maosong Sun

ACL2025 **Oral** OpenCoder: The Open Cookbook for Top-Tier Code Large Language Models

Siming Huang, Tianhao Cheng, Jason Klein Liu, Weidi Xu, JIARAN HAO, Liuyihan Song, Yang Xu, Jian Yang, Jiaheng Liu, Chenchen Zhang, Linzheng Chai, Ruifeng Yuan, Xianzhen Luo, Qiufeng Wang, YuanTao Fan, Qingfu Zhu, Zhaoxiang Zhang, Yang Gao, Jie Fu, Qian Liu, Houyi Li, Ge Zhang, Yuan Qi, Xu Yinghui, Wei Chu, Zili Wang.

ACL2025 (Findings) ChartEdit: How Far Are MLLMs From Automating Chart Analysis? Evaluating MLLMs' Capability via Chart Editing

Xuanle Zhao*, Xuexin Liu*, Yang Haoyue*, Xianzhen Luo, Fanhu Zeng, Jianling Li, Qi Shi, Chi Chen

KDD 2025 Advancing Tool-Augmented Large Language Models via Meta-Verification and Reflection Learning

Zhiyuan Ma, Jiayu Liu, Xianzhen Luo, Zhenya Huang, Qingfu Zhu, Wanxiang Che

EMNLP 2024 Python is Not Always the Best Choice: Embracing Multilingual Program of Thoughts

Xianzhen Luo, Qingfu Zhu, Zhiming Zhang, Libo Qin, Xuanyu Zhang, Qing Yang, Dongliang Xu, Wanxiang Che

EMNLP 2024 Make Some Noise: Unlocking Language Model Parallel Inference Capability through Noisy Training

Yixuan Wang*, Xianzhen Luo*, Fuxuan Wei, Yijun Liu, Qingfu Zhu, Xuanyu Zhang, Qing Yang, Dongliang Xu, Wanxiang Che

COLING 2024 A Survey on Natural Language Processing for Programming

Qingfu Zhu, Xianzhen Luo, Fang Liu, Cuiyun Gao, Wanxiang Che

ACL2022 (Findings) Inverse is Better! Fast and Accurate Prompt for Few-shot Slot Tagging

Yutai Hou, Cheng Chen, Xianzhen Luo, Bohan Li, Wanxiang Che.

AI Open 2022 Augmented and challenging datasets with multi-step reasoning and multi-span questions for Chinese judicial reading comprehension

Qingye Meng, Ziyue Wang, Hang Chen, Xianzhen Luo, Baoxin Wang, Zhipeng Chen, Yiming Cui, Dayong Wu, Zhigang Chen, Shijin Wang

Arxiv CVE-Factory: Scaling Expert-Level Agentic Tasks for Code Security Vulnerability

Xianzhen Luo*, Jingyuan Zhang*, Shiqi Zhou*, Rain Huang*, Chuan Xiao, Qingfu Zhu, Zhiyuan Ma, Xing Yue, Yang Yue, Wencong Zeng, Wanxiang Che

Arxiv **From Code Foundation Models to Agents and Applications: A Practical Guide to Code Intelligence**
Core Contributor

Arxiv **Scaling Laws for Code: A More Data-Hungry Regime**
Xianzhen Luo*, Wenzhen Zheng*, Qingfu Zhu, Rongyi Zhang, Houyi Li, Siming Huang, Yuantao Fan, Wanxiang Che

Arxiv **Step-3 is Large yet Affordable: Model-system Co-design for Cost-effective Decoding**
Core Contributor

Arxiv **Success is in the Details: Evaluate and Enhance Details Sensitivity of Code LLMs through Counterfactuals**
Xianzhen Luo, Qingfu Zhu, Zhiming Zhang, Mingzheng Xu, Tianhao Cheng, Yixuan Wang, Zheng Chu, Shijie Xuyang, Zhiyuan Ma, YuanTao Fan, Wanxiang Che.

Arxiv **Is Compression Really Linear with Code Intelligence?**
Shijie Xuyang*, **Xianzhen Luo***, Tianhao Cheng, Zheng Chu, Houyi Li, Ziqi Wang, Siming Huang, Qingfu Zhu, Qiufeng Wang, Xiangyu Zhang, Shuigeng Zhou, Wanxiang Che.

Arxiv **Automated Snippet-Alignment Data Augmentation for Code Translation**
Zhiming Zhang, Qingfu Zhu, **Xianzhen Luo**, Yixuan Wang, Bohan Li, Wanxiang Che

Arxiv **Format-Adapter: Improving Reasoning Capability of LLMs by Adapting Suitable Format**
Dingzirui Wang, Xuanliang Zhang, Rongyu Cao, Longxu Dou, **Xianzhen Luo**, Yingwei Ma, Qingfu Zhu, Wanxiang Che, Binhua Li, Fei Huang, Yongbin Li

Arxiv **Semi-Instruct: Bridging Natural-Instruct and Self-Instruct for Code Large Language Models**
Xianzhen Luo, Qingfu Zhu, Zhiming Zhang, Xu Wang, Qing Yang, Dongliang Xu, Wanxiang Che

PROJECTS

CVE-Factory: Scaling Expert-Level Agentic Tasks for Code Security Vulnerability Oct 2025 – Present

- A fully automated framework for reproducing CVE vulnerabilities, achieving 95% alignment rate with reproduction from code security experts.
- Released 1000+ executable CVE environments, training data, and Abacus-cve model (32B) that surpasses Qwen3-Coder-480B and Claude Sonnet 4.

Huozi: An Open-Source Universal LLM | 🏆 12 ☆ 200 Mar 2023 – May 2023

- Served as a core developer, responsible for the collection and curation of data for code pretraining and post-training.
- Led the collection and organization of Chinese pretraining datasets.

Abacus: A Lightweight Code LLM | 🏆 2 ☆ 47 Oct 2023 – Sep 2024

- Abacus with 2.7B parameters, outperforms other Code LLMs with parameters $\leq 3B$ such as Stable Code-3B and Granite-3B-Code on both coding and general language tasks.
- Led post-training data construction.

EXPERIENCE

Kuaishou Technology **Beijing, China**
Kstar Research Intern Aug 2025 – Present

- Focus on terminal agent development and research.

StepFun AI. **Beijing, China**
Research Intern Dec 2024 – Jul 2025

- The code aspects of LLM pretraining.
- Developed code data cleaning, training & evaluation pipelines. Provided core code pretraining data for Step3 LLM. Implemented several specialized pretraining tasks/strategies on code.

Du Xiaoman (Beijing) Science Technology Co., Ltd. **Beijing, China**
University-Industry Collaboration Researcher Nov 2023 – Sep 2024

- Explore research to enhance the code generation ability of Code LLMs, leverage code to assist other tasks, and accelerate LLM inference.

Joint Laboratory of HIT and iFLYTEK Research (HFL), China. **Beijing, China**
Research Intern Mar 2022 – Aug 2022

- Focus on Chinese Spell Check.